

Characterization of Preferential Flow Pathways through Paddy Bunds with Dye Tracer Tests

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In terraced paddy landscapes, water losses through the bunds surrounding paddy rice (*Oryza sativa* L.) fields may significantly reduce water use efficiency. The objective of this research was to identify the dominant pathways active for water flow and solute transport in bunds, and to detect differences between the bunds and adjacent field areas. Two earthen bunds aged 20 and 100 yr were investigated by means of dye tracer experiments in Jiangxi Province, China. Most prominently, the bunds were shown to be distinctly more vulnerable to preferential transport of solutes toward the groundwater than the actual paddy fields. The younger bund showed greater water losses than the old one, caused by a greater macroporosity, lower bulk density, and a less effective hard pan. Three successively operating transport processes could be distinguished: (i) dye solution infiltrated vertically into the field and laterally into the bund; (ii) the plow pan underneath the field and the hard pan in the bund then acted as flow barriers, and horizontal spreading occurred on top of the hard pan toward the outer bund and cross-flow to the adjacent field was observed; and (iii) macropore flow occurred through the hard pan and partly in the subsoil down to the groundwater level, initiated on top of the saturated hard pan and routed through earthworm burrows and root channels as well as shrinkage cracks in the younger bund. In conclusion, compaction of a bund may significantly reduce infiltration rates but does not prevent preferential solute losses.

Rice is the major staple food in Asia, where 92% of the world's rice is produced and consumed (IRRI, 1997). Large quantities of water are consumed in paddy rice cultivation, while a decline in freshwater availability is widespread (Li and Barker, 2004).

As a result, approximately 15 million ha of irrigated rice in South and Southeast Asia will suffer physical water scarcity (water demand exceeds available water resources), and 22 million ha will suffer economic water scarcity (insufficient infrastructure or institutional or financial capacity prevents people from accessing water) in 2025 (Tuong and Bouman, 2003; Comprehensive Assessment of Water Management in Agriculture, 2007). For irrigated rice production, large amounts of fertilizers and pesticides are used. As much as 550 to 650 kg N ha⁻¹ yr⁻¹ are applied to two rice crops in eastern China (Zhu et al., 2003). Bypassing the biologically active soil matrix in macropores reduces the time for sorption and degradation of the fertilizers and pesticides added and may endanger groundwater resources, which are often shallow and used for drinking water purposes.

It has been demonstrated that the bunds (also referred to as levees, banks, or dikes) surrounding paddy rice fields are responsible for a large part of the total water losses. These water losses through bunds have been determined by comparing total per-

colation losses with infiltration rates measured in the field area (Tuong et al., 1994; Kukal and Aggarwal, 2002; Bouman et al., 2005). Water that infiltrates the bund may either percolate vertically toward the groundwater or flow through the bund into the adjacent field as cross-flow (Fig. 1). The vertical underbund percolation is also named downward flow type (Huang et al., 2003; Tsubo et al., 2007) or lateral percolation (Walker and Rushton, 1984, 1986), while cross-flow is also referred to as horizontal flow type (Huang et al., 2003) or lateral seepage (Tsubo et al., 2007).

For land preparation, paddy rice fields are flooded and subsequently puddled. Puddling comprises repeated plowing, harrowing, and finally leveling the soil surface (Wopereis et al., 1992). This causes the breakdown of aggregates and the destruction of macropores (Ringrose-Voase et al., 2000), and induces the formation of a plow pan, minimizing percolation losses (e.g., Kukal and Aggarwal, 2002; Arora et al., 2006). If a bund within the field is shifted to a new location, or if a terraced bund slowly moves backward with time, this plow pan may extend underneath the bund, which is of great significance for water fluxes (Huang et al., 2003). We refer to a compacted layer within the bund with the term *hard pan*, which is not confined to a specific origin (see also Fig. 1). Generally, hydraulic properties and accordant water movement have been assumed to be uniform throughout the investigated bunds in hydrologic modeling (Odhiambo and Murty, 1996; Walker and Rushton, 1984).

Preferential flow can be defined as water and solutes moving along certain pathways while bypassing a fraction of the porous soil matrix, and comprises macropore, unstable, and funnel flow (Hendrickx and Flury, 2001). Macropore flow occurs mainly in fine-textured soils with a pronounced structure, and ponding water increases the risk for macropore flow (Flury et al., 1994; Hendrickx and Flury, 2001; Clothier et al., 2008). Both fine soil texture and continuous ponding

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conditions are typical for paddy rice fields. Macropores develop as root channels (Mori et al., 1999), earthworm burrows (Sander and Gerke, 2007), and shrinkage cracks (Ringrose-Voase et al., 2000) in the puddled layer as well as in the plow pan. In continuously submerged fields, cracks form after final drainage. When water scarcity occurs or when intermittent irrigation is adopted, however, cracks may also form during the rice growing season and attain maximum aperture under extended droughts. Not only do cracks enhance vertical percolation, but they also act as horizontal feeders to the highly permeable bunds (Janssen and Lennartz, 2007).

For visualization, identification, and quantification of water flow pathways in structured soils, dye tracer experiments are a common method (e.g., Flury et al., 1994; Stadler et al., 2000; Ohrström et al., 2004). Brilliant Blue FCF has been widely used for this purpose, as it is nontoxic, is distinctly visible in soils, and can be considered moderately mobile at large concentrations (Flury and Flühler, 1995; Kasteel et al., 2002; Mon et al., 2006). It must be noted, however, that its sorption isotherm is nonlinear and depends on soil properties (Ketelsen and Meyer-Windel, 1999; Perillo et al., 1999; German-Heins and Flury, 2000). By means of dye tracer experiments, Yasuda et al. (2001) found that, for a heavy clay soil, water infiltrated predominantly along shrinkage cracks under ponding conditions. Wopereis et al. (1994) demonstrated that cracks reached beyond the root zone down to 65-cm depth in a rice field of silty clay loam. Within our research catchment, dye tracers have been used for the analysis of soil structure within the puddled part of the paddy fields (Sander and Gerke, 2007). These researchers could demonstrate preferential water fluxes through the plow pan along biopores and shrinkage cracks and showed that puddling did not fully destroy the connectivity between macropores. Their study, however, as well as other investigations, concentrated on the field area. It remains largely unknown which transport processes operate in the bunds.

The objectives of this study were to assess if bunds are more vulnerable to preferential solute movement than the adjacent paddy rice fields and to describe the dominant flow pathways in terraced paddy bunds. Two earthen bunds, 20 and 100 yr old, were therefore investigated by means of dye tracer experiments.

MATERIALS AND METHODS

Experimental Sites

The experiments were conducted in the Sun Jia catchment, which is situated in the vicinity of the Ecological Experimental Research Station of Red Soil (116.90° E, 28.23° N), Jiangxi Province, in southeastern China. The region is characterized by low hills, and the paddy rice fields are arranged in terraces with a slope of 2 to 5%. The average field size is 500 m². The soils are classified as Hydragric Anthrosols according to the World Reference Base (FAO, 2006) and as Plinthudults in the U.S. Soil Taxonomy (Soil Survey Staff, 1999). The soils developed from Quaternary red clays and comprise predominantly loam and clay loam. Two rice crops are grown every year, from April to July and from August to October or November, while green manure dominates during the winter fallow. Land preparation starts in April with puddling of the soil, which is done with the help of water buffalo (*Bubalus bubalis*). Continuous ponding conditions are then maintained until 2 wk before harvesting the first rice crop. During the second rice crop, seasonal droughts may occur, causing periodic

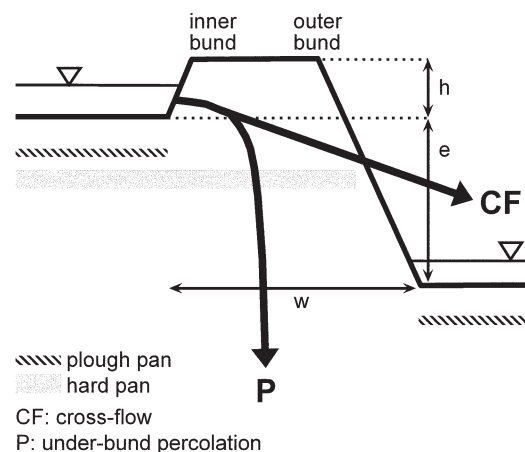


Fig. 1. Sketch of a bund between terraced paddy fields (not to scale); h is height, e is elevation difference, w is width.

drying of the fields, as half of the total annual rainfall of 1706 mm falls from March to July (Zhang et al., 2004). The experiments were conducted after rice harvest in November and December 2005.

Soil and Bund Properties

Bunds are constructed manually with soil from the neighboring fields in the investigated catchment as well as in large parts of China. Soybean [*Glycine max* (L.) Merr.] was grown on most bunds, which had already been harvested when the experiments were conducted. Two bunds of different age, which are typical for the catchment, were chosen for the experiments. Bund BY is situated on the downward side of Field QY and was constructed 20 yr ago, when a tea [*Camellia sinensis* (L.) Kuntze] plantation was converted into paddy fields. The elevation difference e between the adjacent fields is 42 cm, the base width w of the bund is 114 cm, and its height h is 17 cm (see also Fig. 1). Bund BO is located on the downward side of Field QO, which has been used for paddy cultivation for >100 yr. It is slightly larger than the younger bund, with 62-cm difference in elevation, a base width w of 132 cm and a height h of 12 cm. Bund BO has been intensively used as a pathway for people and cattle for several decades. In both Fields QY and QO, typical paddy soil profiles have developed, with a puddled layer overlying a distinct plow pan (Table 1; Janssen and Lennartz, 2007).

The soil texture of Bund BY is clay loam throughout the profile, while Bund BO is loam from the top down to the 25-cm depth, with clay loam underneath. The soil color in Bund BY is 5YR 4/4 down to the 35-cm depth and 2.5YR 4/6 underneath. Soil color in Bund BO is 10YR 4/3 down to the 19-cm depth, 7.5YR 5/6 from 19 to 30 cm, 7.5YR 3/4 from 30 to 35 cm, and 10YR 4/6 underneath. Bulk density,

Table 1. Soil properties of Fields QY and QO. Bulk densities are arithmetic means, hydraulic conductivities are geometric means. Standard deviations are given in parentheses.

Field	Depth	Texture	Bulk density	Hydraulic conductivity
	cm		g cm ⁻³	cm d ⁻¹
QY	0–14	clay loam	1.29 (0.04)	42 (97)
	14–18	clay loam	1.56 (0.03)	0.79 (1.14)†
	>18	silty clay	1.26 (0.05)	8.3 (5.1) ‡
QO	0–14	loam	1.00 (0.05)	76 (95)
	14–19	loam	1.70 (0.02)	0.16 (0.17) †
	19–29	loam		
	>29	clay loam	1.52 (0.04)‡	5.0 (6.6)‡

† Infiltration rate measured with double-ring infiltrometers.

‡ Data from Janssen (2008).

Table 2. Bulk density (arithmetic means) and saturated hydraulic conductivity (geometric means) of Bunds BY and BO. Standard deviations are given in parentheses.

Depth cm	Bulk density g cm ⁻³	Saturated hydraulic conductivity cm d ⁻¹
<u>Bund BY</u>		
-10	1.32 (0.08)	30.4 (114.1)
0	1.31 (0.03)	502.2 (308.5)
10	1.37 (0.05)	36.9 (1426.2)
20	1.46 (0.06)	113.2 (818.5)
30	1.47 (0.02)	13.3 (77.0)
40	1.39 (0.04)	7.0 (15.1)
60	1.33 (0.04)	2.2 (7.0)
<u>Bund BO</u>		
0	1.48 (0.07)	37.4 (144.6)
12	1.54 (0.05)	66.8 (241.2)
18	1.65 (0.05)	6.3 (31.3)
33	1.62 (0.04)	
40	1.51 (0.06)	5.5 (5.2)
70	1.50 (0.04)	11.6 (43.6)

measured with 100-cm³ steel cylinders at six (BY) and eight (BO) replicates per depth, was distinctly higher in Bund BO than in Bund BY and revealed a hard pan in both bunds (Table 2). Hydraulic conductivities, determined with the constant-head method in the same cylinders, were extremely high and variable in both bunds. The geometric mean ranged from 2.2 cm d⁻¹ in the subsoil to 502.2 cm d⁻¹ at field level in Bund BY (Table 2).

Dye Tracer Experiments

On each of the investigated bunds, BY and BO, two dye tracer experiments were performed. The experimental sites BY1 and BY2 were located 5.5 m apart, and BO1 and BO2 8.5 m apart from each other. For each experiment, a three-sided metal frame sized 1.2 by 0.8 m with a height of 0.3 m was carefully inserted into the bund (Fig. 2). The frame was driven into the field's soil approximately 16 cm deep, down into the plow pan, and it reached 0.4 m into the bund. Minor fissures between frame and soil were sealed with wet soil to prevent preferential movement of the dye solution. Field and bund surfaces were kept intact, and four rice stubbles were enclosed by the frame. Initial pressure heads in the bunds were determined for Exp. BY1 and BO1 with tensiometers outside the experimental plots at four depths and two positions, and the groundwater level was registered.

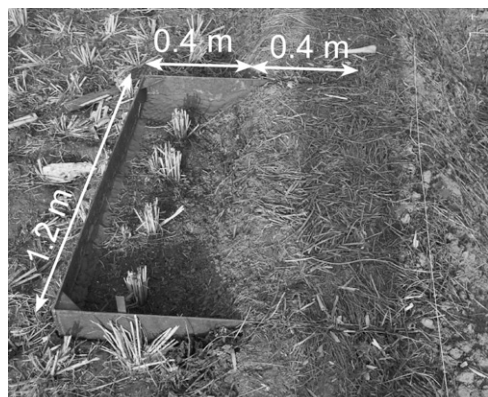


Fig. 2. Three-sided metal frame used for dye infiltration.

A 5-cm water pulse containing 7.5 g L⁻¹ Brilliant Blue FCF was poured onto a plastic sheet within the frame. The plastic sheet was then carefully removed. This procedure ensured that flooding conditions were identical in all experiments. Subsequently, the water level was kept constant at 5 cm with another 5 cm of dye solution, resulting in a total of 10 cm of applied Brilliant Blue solution (48 L), to simulate ponding conditions typical for the investigated catchment. The infiltration rate was measured.

The following day, one half of the experimental plot was dug up in about 20 horizontal cross-sections (19 for BY2, 10 for BO2), and the other half in about 10 vertical profiles (11 for BY1, 7 for BO1) (Fig. 3), all of which were carefully prepared with a spatula. Horizontal cross-sections allow a better assessment of the macropore distribution, while vertical profiles depict the water flow pathways in the profile. A combination of both is an established method (Flury et al., 1994; Stadler et al., 2000; Yasuda et al., 2001; Kim et al., 2004) and allows a cross-check (Sander and Gerke, 2007). In general, horizontal cross-sections were prepared every 2.5 cm down to the 10-cm depth, every 5 cm to the 40-cm depth, and every 10 cm to the 90-cm depth, although this scheme was modified when necessary to account for varying soil profiles. Spacing between vertical profiles was 5 to 6 cm. A border of 3 cm along the frame was not taken into account for image analysis. Resulting horizontal profiles had a dimension of 54 by ≤157 cm, while vertical profiles reached down to 82- to 85-cm depth and were also up to 157 cm wide.

Image Processing

Photos were taken with a Canon EOS 33 camera and a Canon 28- to 90-mm lens on 200 ASA Kodak Elite Chrome slide film. A rectangular frame was placed around both horizontal and vertical profiles, and direct sunlight was avoided by erecting a shade. For each horizontal cross-section, two photos were necessary to cover the entire length of the section. Each of these images was analyzed independently. They were only merged after final classification. Slides were digitized with a Nexscan F4100 scanner at a spatial resolution of 4000 dpi and a spectral resolution of 16 bits.

For geometric correction, a first-order polynomial was fitted to selected control points (i.e., points with known coordinates) by least squares. It was chosen to avoid distortions within the image, as control points were only located along the border of the image (Richards and Jia, 2006). Nearest neighbor resampling was chosen to keep the original values for subsequent image analysis. The pixel size was set at 0.02 cm for horizontal cross-sections and 0.04 cm for vertical profiles.

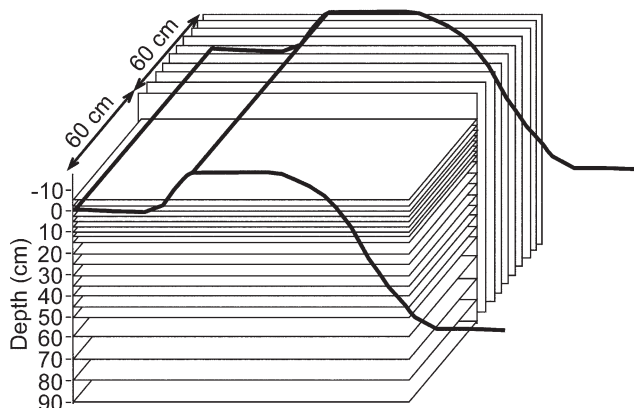


Fig. 3. Sketch of analyzed profiles.

Results achieved from an illumination correction by background subtraction were not satisfactory. Instead, a band ratio was used for image classification. Band ratioing is commonly used in remote sensing and compensates for brightness variations, as illumination increases the digital numbers in both bands, and will thus disappear in the ratioed image (Lillesand et al., 2004). The blue and red bands were chosen for ratioing because they were least correlated. The blue band showed the stained soil pattern and the red one the soil background. The band ratio was calculated by dividing the digital number in the blue band by the corresponding value in the red band for each pixel in the image. For our experiments, the ratioed image allowed a much better discrimination of stained and unstained areas than the cyan channel (after transformation from the RGB to the CMYK color space) as proposed by Baveye et al. (1998) and Ohrström et al. (2004). We visually determined a threshold to discriminate between stained and unstained areas in the band ratio image. This step was undertaken using the dynamic overlay function of the ENVI 3.4 image processing software (Research Systems, Boulder, CO), which allowed an immediate comparison of the classified image and the color photograph. This procedure included a subjective choice in threshold determination, but automatic thresholding by means of the Intermeans algorithm (Baveye et al., 1998) did not yield satisfactory results.

After classification into a binary image, a 3 by 3 median filter was applied to remove a salt and pepper structure and to smooth contours, followed by removal of all objects $<2 \text{ mm}^2$ to increase the legibility of the images.

Generation of Dye Coverage Diagrams

The results of dye tracer experiments are usually reported as graphs depicting average dye coverage, which is a binary rating as stained or unstained, vs. depth (e.g., Flury et al., 1994). Our study, however, was focused on two-dimensional flow processes. We therefore created graphics depicting the average dye coverage as a function of depth and horizontal distance. The average dye coverage in the vertical profiles was calculated by averaging all binary images of one experiment (Kulli et al., 2003). The graphic displaying the horizontal cross-sections was generated as follows (Fig. 4): For each row of pixels in the horizontal cross-section (corresponding to 54 cm or 2700 pixels), the average dye coverage was calculated and converted to a gray-scale value. The resulting sequence was then extended from the cross-section's depth to the adjacent cross-sections.

For further analyses, we divided the experimental profiles into three regions: the paddy field at the left rim of the profiles (0–25 cm), a transition zone between paddy field and bund (25–50 cm), and the bund itself (>50 cm). The maximum dye extent in the bund was at a distance of 130, 153, 157, and 148 cm in Exp. BY1, BY2, BO1, and BO2, respectively. For reasons of comparison, dye coverage in all bunds was calculated with a standardized bund width of 157 cm.

RESULTS AND DISCUSSION

Infiltration Rates

Average infiltration rates during the experiments were 31,447 (BY1), 42,647 (BY2), 1433 (BO1), and 1549 $\text{cm}^2 \text{d}^{-1}$ (BO2). The unit square centimeters per day resulted from the volume of infiltrated water per unit bund length and unit time. The former infiltration rates were extremely high, while the latter ones lay within the range reported previously (Tuong et al., 1994; Walker and Rushton, 1984; Kukal and Aggarwal, 2002). Cross-flow occurred during Exp. BY2, BO1, and BO2.

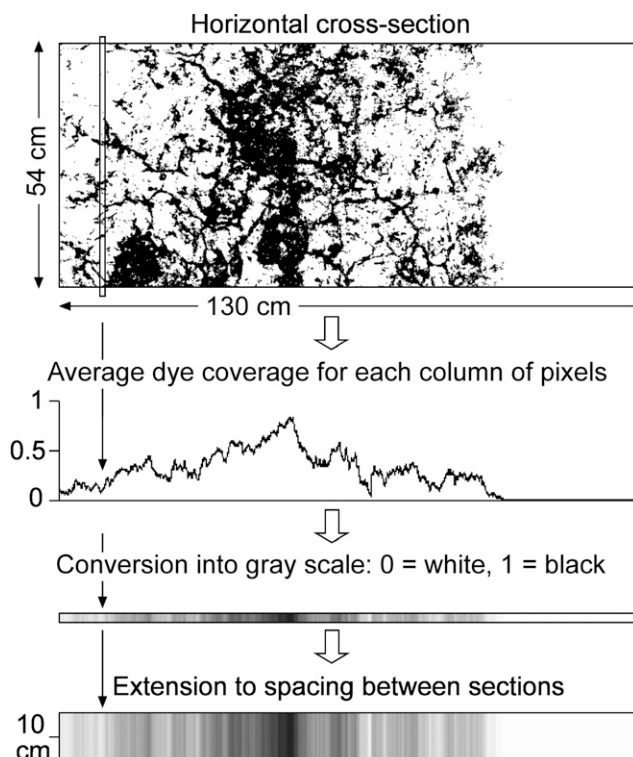


Fig. 4. Preparation of dye coverage profiles out of horizontal cross-sections.

No leakage of dye solution along the frame's side walls was observed; however, some lateral spreading occurred in the bund beyond the extent of the frame. The different infiltration rates at Bunds BY and BO caused different experiment durations. The first 5 cm of water took 9.5, 6, 210, and 193 min to infiltrate at Exp. BY1, BY2, BO1, and BO2, respectively. Pressure heads before dye infiltration were higher in Bund BO than in Bund BY (Fig. 5), which may partly explain the lower infiltration rates. In the fields, average pressure heads at 15- and 25-cm depth were -15.1 and -9.7 kPa at Exp. BY1, and -2.9 and -2.4 kPa at Exp. BO1. The differences were caused by a generally diverging moisture regime of Field QO, which drains slowly according to local farmers, as a result of a low-permeability plow pan and limited bund infiltration.

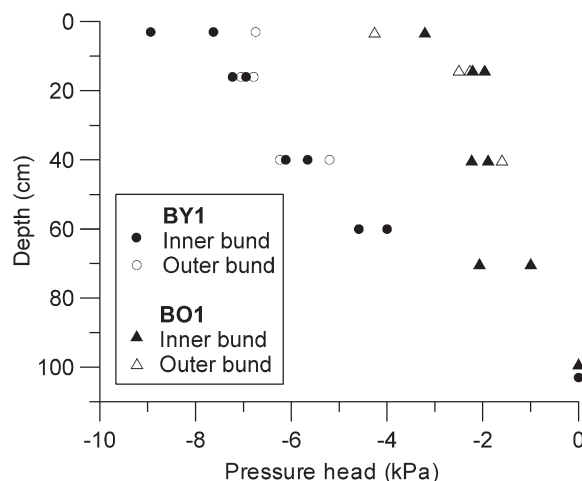


Fig. 5. Initial pressure heads at Exp. BY1 and BO1.

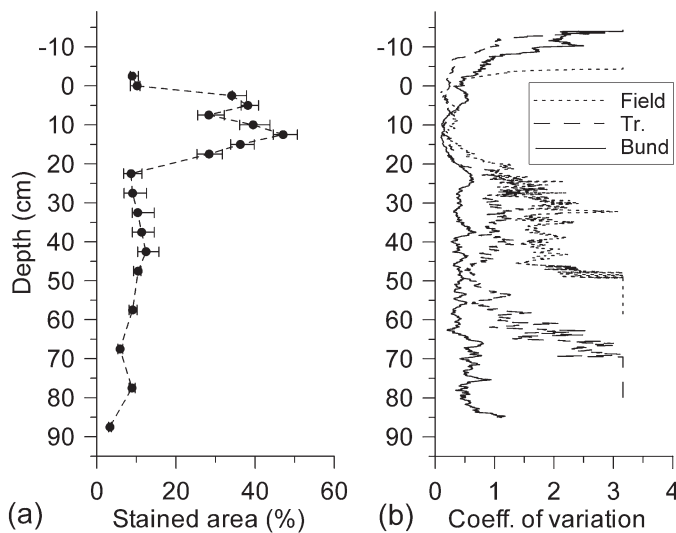


Fig. 6. Experiment BY2: (a) average dye coverage with classification errors of total horizontal cross-sections, and (b) coefficients of variation for vertical profiles. Tr is the transition zone.

Quality of Image Classifications

Before further analysis of the data, the goodness of the image classification results was tested, as thresholding of images is subjective and may lead to biases (Baveye et al., 1998). Experiment BY2 was chosen arbitrarily for this purpose. Repeated, inde-

pendent classifications of the same image showed that a chosen threshold in the band ratio image (to distinguish between stained and unstained pixels) could be determined reliably within an error interval of 0.025. Therefore, for each horizontal cross-section, the classification procedure was repeated twice by adding and subtracting 0.025 to the chosen threshold, and the respective stained area was calculated (see error bars in Fig. 6a). Figure 6a shows that the error range was comparably small and amounted to less than $\pm 4\%$ of the stained area. Band ratioing with visual threshold determination is therefore a suitable method for classification of dye tracer images. It is only reliable, however, if the background color of the soil is uniform throughout the profile. In the vertical profiles, especially at experimental site BO, soil color changed with depth; therefore, the dye coverage may be slightly overestimated in the upper part and underestimated in the lower part of the profile.

Horizontal Dye Patterns

Selected processed horizontal cross-sections of Bunds BY and BO are presented in Fig. 7.

In Exp. BY1, the upper bund down to 13-cm depth was characterized by a high dye coverage (Fig. 7a), and interaction between matrix and macropores (cracks and biopores) was high. The dye coverage abruptly decreased underneath, and dye was concentrated in the remains of rice stubbles connected by cracks. At 20- to 25-cm depth (the upper hard pan), a crack

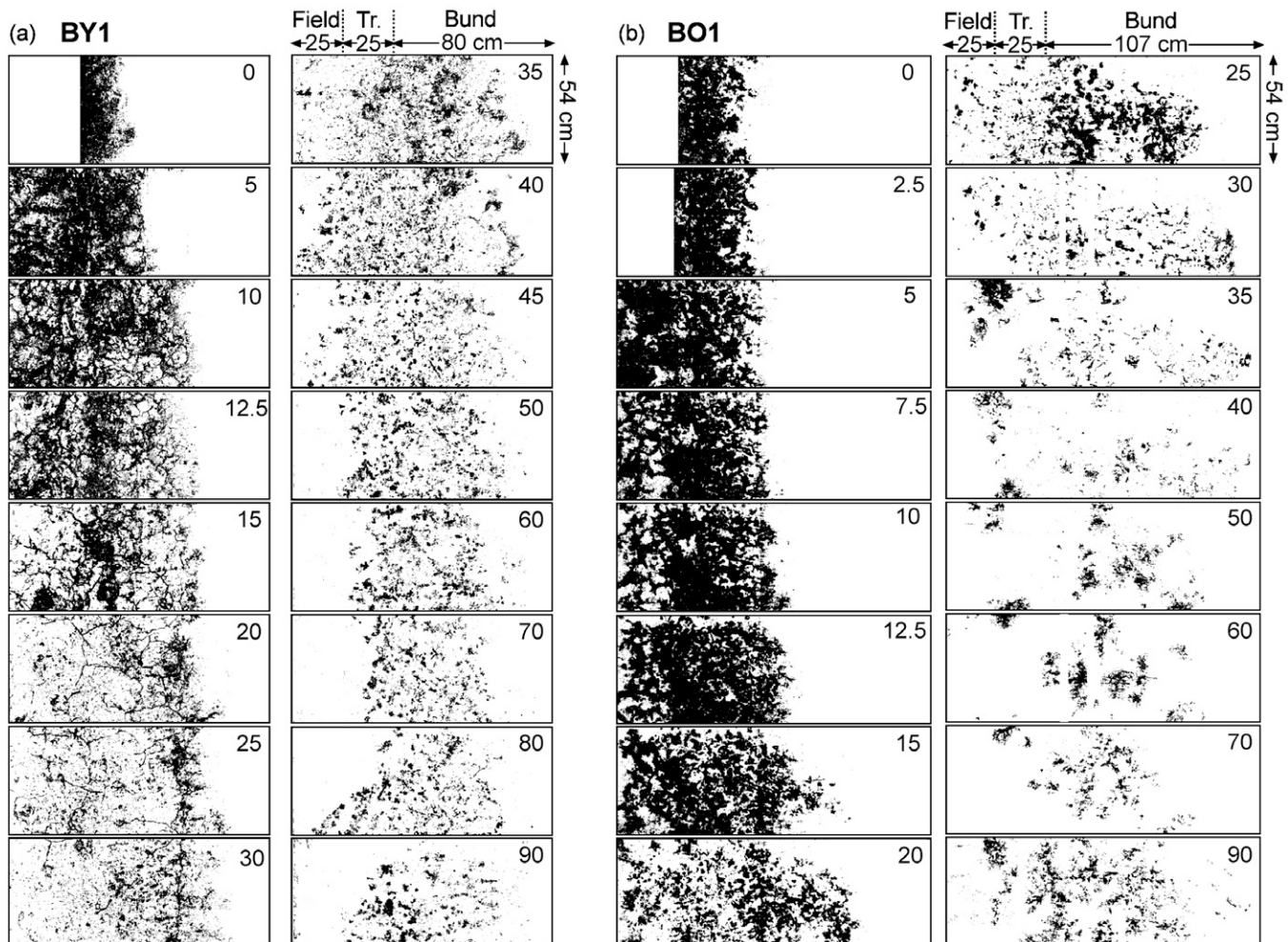


Fig. 7. Selected classifications of horizontal cross-sections for (a) Exp. BY1 and (b) Exp. BO1. The paddy field, from where the dye was infiltrated, is located on the left-hand side of the images. Numbers refer to depths (in centimeters). Tr is the transition zone.

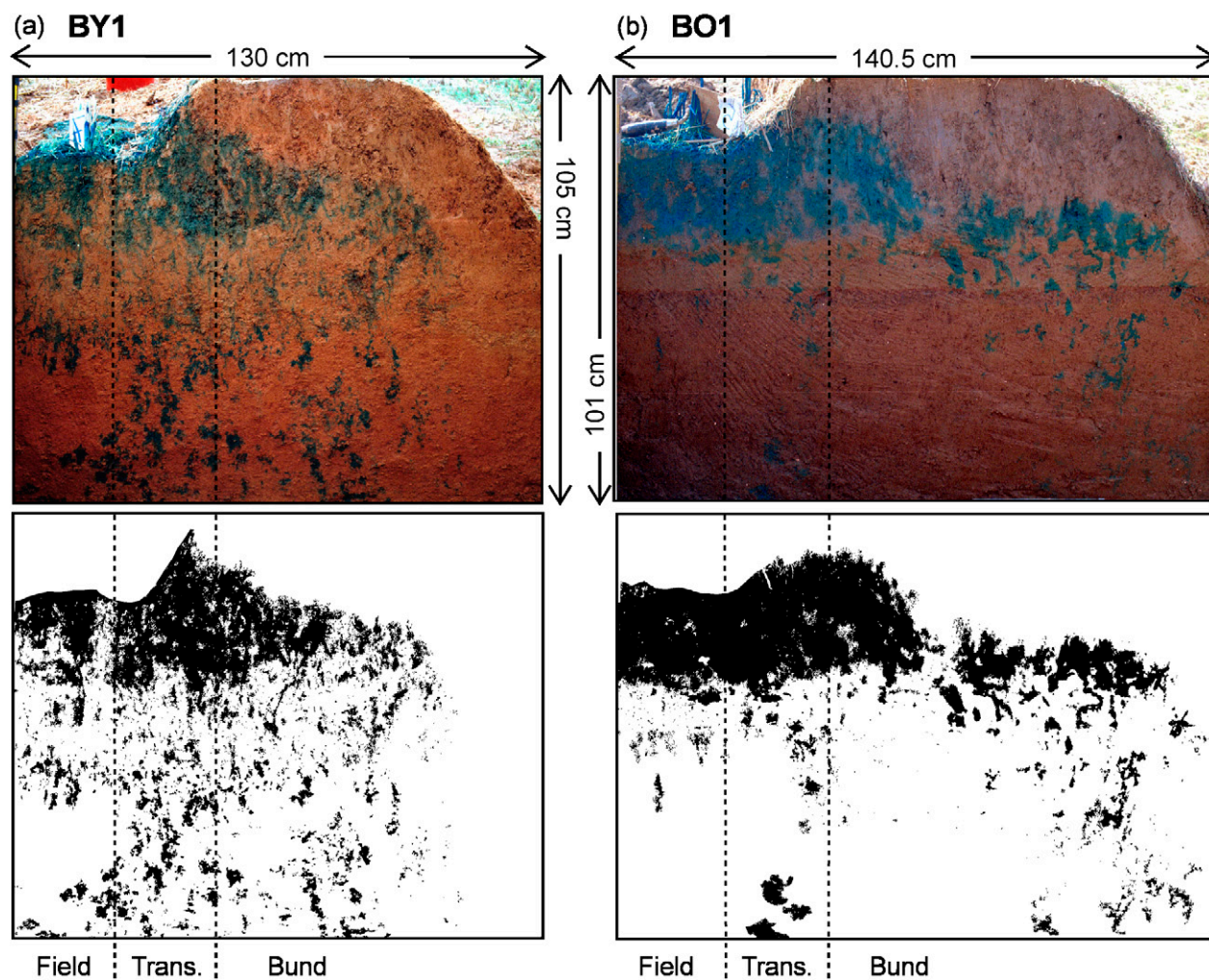


Fig. 8. Selected vertical profiles with accordant classifications of (a) Exp. BY1 and (b) Exp. BO1. Trans. = transition zone.

network dominated the dye pattern while most of the matrix was bypassed. The cracks were present in the transition zone and reached far into the bund, while they were less pronounced in the field area. This distribution indicates that cracks form at first in the vicinity of the bund parallel to its orientation, when the bund has already dried out but the water content within the field is still high. Upon rewetting, cracks may then enhance infiltration into the bund. Underneath at a depth of 30 and 35 cm (the lower hard pan), stained earthworm burrows as well as root channels with a diameter of around 0.5 mm, possibly from soybean, were visible. Cracks and biopores were surrounded only by a thin stained ring, indicating a limited interaction between macropores and the surrounding soil matrix. Vertical profiles (Fig. 8a) confirmed that most of the matrix was bypassed in the hard pan between 20- and 35-cm depth, where dye could only be found in narrow vertical channels. From the 40-cm depth down to the bottom of the profile, along with a decrease in bulk density, stained spots with a diameter of a few centimeters dominated and the dye patterns remained spatially similar with depth. The spots were generally not associated with macropores and may reflect matrix-driven fluxes underneath macropores. This phenomenon was also observed by Yasuda et al. (2001), where dye was mainly transported in cracks down to 0.65-m depth, while a textural and structural change enhanced dispersion and a more homo-

geneous flow field underneath. When comparing the horizontal dye distribution from the top to the bottom of the profile (Fig. 7a), a distinct shift from the field area toward the outer bund can be observed. It demonstrates the relative importance of underbund percolation compared with field percolation for solute transport to the groundwater. The dye patterns in Exp. BY2 were basically the same as in BY1.

Dye patterns at the older Bund BO were different from the results of Bund BY (Fig. 7b). One important difference was the lack of cracks. It is possible that a more effective hard pan as well as the low-permeability bund caused higher water contents in Field QO and Bund BO. Furthermore, a larger part of the matrix was stained down to about 18-cm depth as a result of reduced macroporosity and dominant matrix fluxes. Stained spots underneath the hard pan were larger than at Bund BY.

Burrows in both bunds originated mainly from earthworms, but also from mole crickets (*Gryllotalpa africana*). Cricket burrows are generally larger than earthworm burrows, with a diameter up to 1.5 cm, but no reliable distinction between burrows of either origin was possible. Burrows were counted manually in the images of both bunds; they were distinguished from root channels by a size >0.4 cm. Within the field area, the burrow density in all experiments was highest at a depth of about 20 cm, corresponding to the plow pan, above which the burrows had been destroyed by puddling.

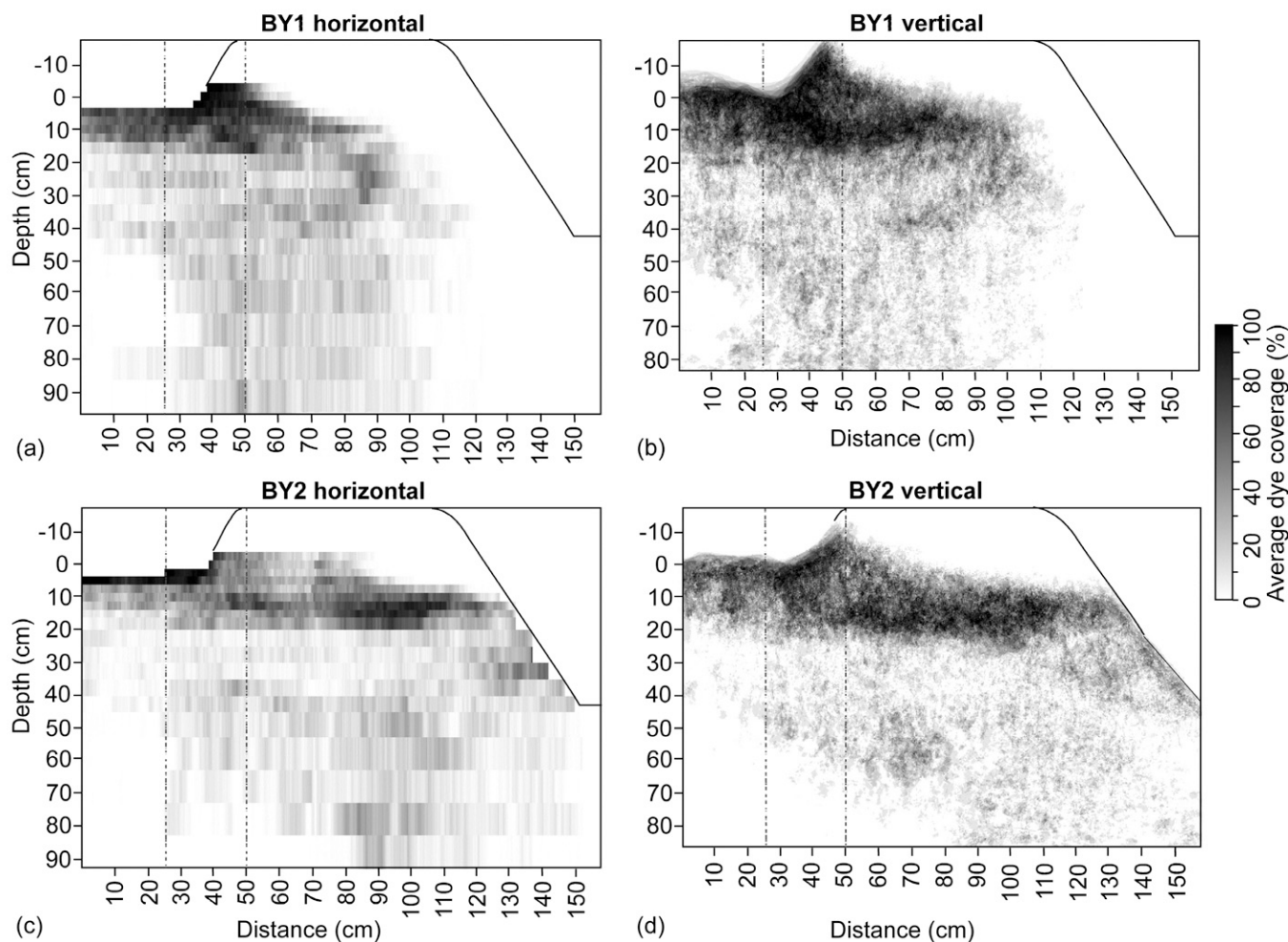


Fig. 9. Average dye coverage of Bund BY: (a) horizontal cross-sections converted to a vertical profile, Exp. BY1; (b) vertical profiles, Exp. BY1; (c) horizontal cross-sections converted to a vertical profile, Exp. BY2; (d) vertical profiles, Exp. BY2. Dashed lines delineate field, transition, and bund regions.

The density ranged from 22 (BO2) to 37 m^{-2} (BY1). In the bunds, however, the density was highest between 5- and 20-cm depth on top of the hard pan (except for the outermost 25 cm), and maximum values ranged from 67 (BO2) to 104 m^{-2} (BO1, BY2). The burrow density in the bunds was generally two- to fourfold higher than in the field area. Biological activity is probably especially high in the bunds, where animals may escape from the water after flooding and migrate back when the field is drained. Mole crickets also lay their eggs in the bunds (IRRI, 1983). Burrows also occurred in the horizontal direction and may enhance lateral water movement on top of the hard pan. The most prominent finding, however, was that nearly all earthworm and cricket burrows were stained in the field area, while the fraction of stained macropores, especially in Bund BO, decreased from the inner to the outer bund, and beyond 1-m distance almost no burrows conducted dye. This implicates that most of the macropores within the field region were already active in transporting dye during the experiments. In the bunds and their hard pans, however, the fraction of operational macropores will increase with the duration of flooding, which possibly further enhances the discrepancies between field area and bund water and solute fluxes.

Dye Coverage

Two-dimensional profiles of the average dye coverage derived from horizontal as well as vertical profiles illustrate the water flow pathways that were active for solute transport. Both averaged images of Exp. BY1 (Fig. 9a and b) show that dye was concentrated in the upper part of the profile, with a rather uniform distribution and a dye coverage of up to 80% down to a depth of about 13 cm below the ground surface of the field area and 17 cm below the bund. These boundaries of dye coverage correspond to the upper limits of the plow pan beneath the field area and of the hard pan in the bund. All images in Fig. 9 suggest that, at first, the dye solution infiltrated vertically into the field and laterally into the bund, resulting in a nearly continuous staining of the upper layer. Below the plow or hard pan, dye coverage decreased abruptly to <20% (Fig. 10a). The compacted layers acted as flow barriers and prohibited a further vertical propagation of the dye solution. Instead, as a second process after infiltration, horizontal spreading occurred on top of the hard pan toward the outer bund. Lateral spreading may have been facilitated by horizontal cracks, which often form between the puddled topsoil and the plow pan during drying (Wopereis et al., 1994; Sander and Gerke, 2007), and by earthworm and cricket burrows. In Exp. BY1, these lateral fluxes ceased at a distance of 90 to 100 cm, and the dye solution was

routed vertically downward. The resulting enhanced dye coverage was also visible in the horizontal cross-sections and was associated with a fissure parallel to the bund (Fig. 7, 15–35-cm depth). It is most likely that the hard pan was not operational or did not exist beyond this distance, which might be a consequence of the farmers' practice to dig up the top outer bund after harvest to regulate mole crickets. Virtually no dye penetrated deeper than 50 cm in the field area, suggesting that the plow pan is efficient in preventing percolation losses to the subsoil. In the bund, however, the dye coverage remained constant at 6 to 8% from 42- to 90-cm depth. It was rather discontinuous within a specific depth as a consequence of preferential flow pathways, which partly reached down to the base of the profile. Macropore flow through the hard pan and to a minor degree through the subsoil was thus the third important process governing solute transport in paddy bunds.

In Exp. BY2 (Fig. 9c, 9d, and 10b), the horizontal fluxes on top of the hard pan continued until the outer border of the bund, and dye coverage throughout the profile was reduced compared with BY1. About 5 to 10 L of dye solution was leaking out of the bund as cross-flow toward the adjacent, lower field and was thus not available for staining the soil. The evidence of cross-flow contradicts simulation results for terraced paddies from Walker and Rushton (1986) and Huang et al. (2003), where no seepage face developed on the surface of the bund.

At Bund BO, the dye coverage in the upper part of the bund was more uniform and elevated (80–90%) compared with Bund BY, which is a consequence of dominating matrix fluxes (Fig. 11, 12, and 8b). The gradient of dye coverage between the topsoil and the subsoil was markedly more pronounced, and shows that the well-developed plow pan and hard pan were more effective in preventing percolation losses. Still, water bypassed the low-permeability layers in vertical biopores. Dye coverage was distinctly lower below the field than below the bund in all subplots, again demonstrating the bunds' importance for total water losses. The larger size of stained spots in the subsoil of Bund BO (compared with Bund BY) was probably caused by a longer experiment duration and an accordingly greater amount of dye transported via macropores to the subsoil.

Dye coverage obtained from horizontal and vertical cross-sections generally showed very good agreement (Fig. 10 and 12), demonstrating that the flow patterns were consistent within one dye experiment. In the upper part of the bund regions of Exp. BY1 and BO2, though, dye coverage was higher in the vertical profiles, where the dye had penetrated farther into the bund. The standard deviation of dye coverage in the vertical profiles was similar in all experiments and compared with the study of Ohrström et al. (2004), in which it was <10% for eight profiles (dye coverage is given in percentage and standard deviation accordingly). For BY2, the standard deviation rose from 6% in the upper bund to 11% at 13-cm depth and then decreased to 1 to 4% below the 16-cm depth. The coefficient of variation in all regions was close to zero from 0- to 10- to 20-cm depth, and then increased toward the bottom of the profile (Fig. 6b). In the bund, it remained around 0.5, while

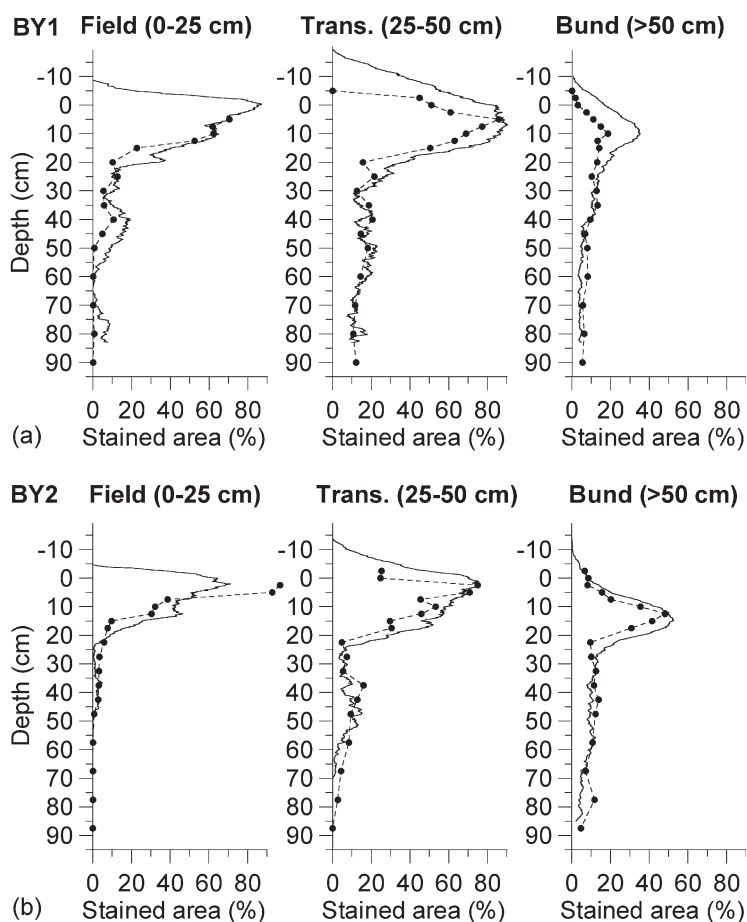


Fig. 10. Average dye coverage of Bund BY as divided into field, transition, and bund regions: (a) Exp. BY1; (b) Exp. BY2. Lines are the vertical profiles, dots are the horizontal cross-sections.

it was distinctly higher in the field areas, with values from 1.5 to 3, showing a higher spatial variability of dye coverage in the field than in the bund. It can be deduced that preferential flow pathways are rather evenly distributed in the bunds, while they occur only sporadically within the fields.

Following the definition of flow regimes by Flüher et al. (1996), the upper part of both bunds corresponds to the attractor zone with distribution flow, while the hard pan acts as a transition zone with dominant preferential flow. A dispersion zone follows underneath in Bund BO. Both prerequisites for macropore flow postulated by Germann and Beven (1981) are fulfilled on top of the hard pan when paddy fields are flooded: The micropore system has to be close to saturation (below -1.0 kPa according to Jarvis, 2007), and vertical flow velocities have to exceed the infiltration capacity of the soil matrix at a permeability break within the soil. Water pressures on top of the hard pan in both Bund BY and Bund BO were near zero or positive when the paddy fields were flooded (data not shown). In addition, the lateral flux rate (of the water infiltrated into the bund and redirected horizontally toward the outer bund) exceeded the infiltration capacity of the hard pan's soil matrix. This subsurface initiation process of macropore flow has been observed before (e.g., Weiler and Naef, 2003; Forrer et al., 1999; Petersen et al., 2001), but it seems particularly important in the flooded paddy bunds, where saturated conditions will remain for long periods of the year. Macropores were

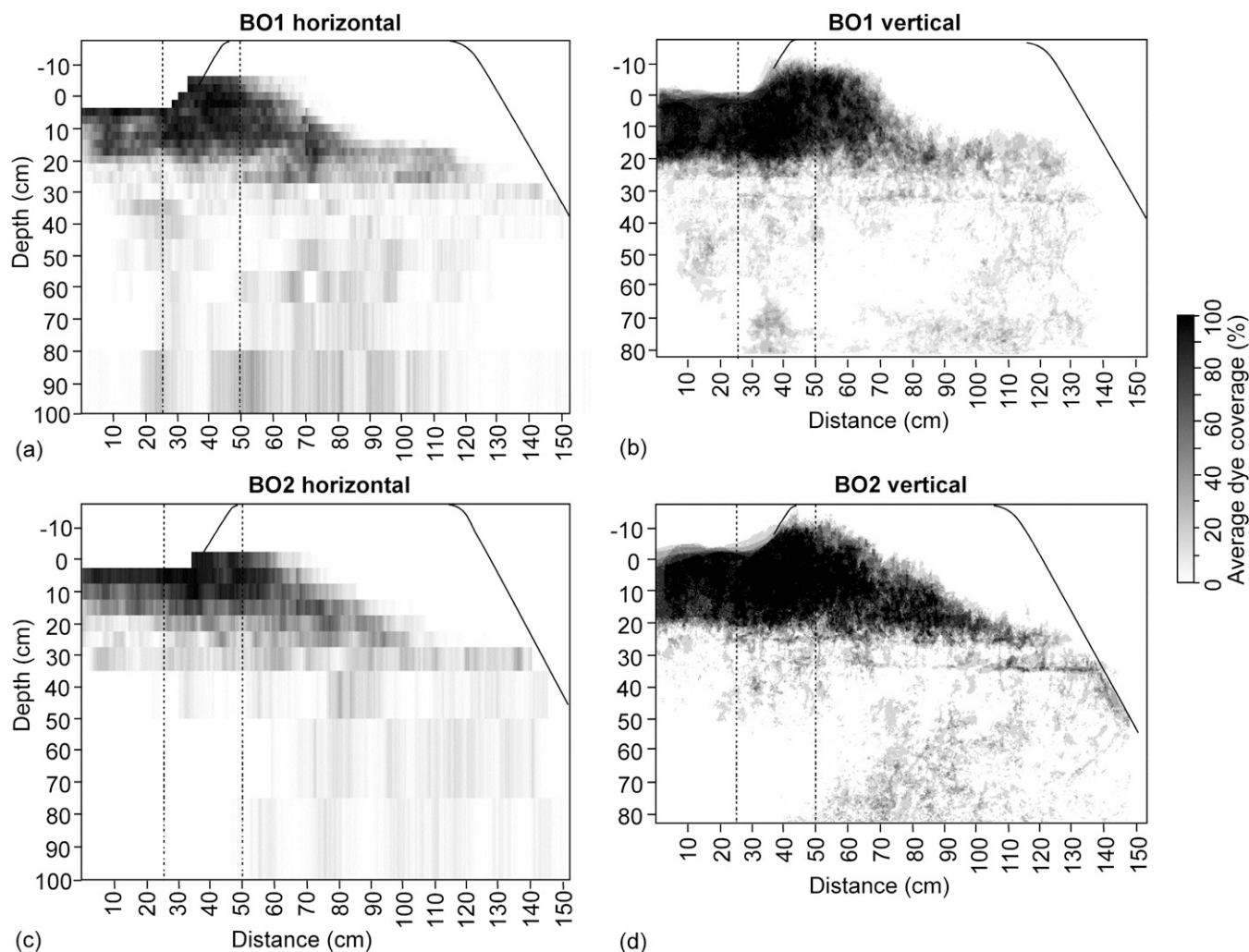


Fig. 11. Average dye coverage of Bund BO: (a) horizontal cross-sections converted to a vertical profile, Exp. BO1; (b) vertical profiles, Exp. BO1; (c) horizontal cross-sections converted to a vertical profile, Exp. BO2; (d) vertical profiles, Exp. BO2. Dashed lines delineate field, transition, and bund regions.

continuous through the hard pan (Fig. 8), which is of great importance despite the layer's small extent. An intact hard pan minimizes underbund percolation and solute leaching, but macropores distinctly increase its saturated hydraulic conductivity (Cameira et al., 2003). Lateral exchange between macropores and matrix is limited in the hard pan, further enhancing preferential flow (Alaoui and Goetz, 2008). Macropore flow has been demonstrated before to be the dominant flow mechanism in specific horizons of various soil types (Weiler and Flühler, 2004; Flury et al., 1994; Petersen et al., 2001; Kim et al., 2004). Despite the occurrence of macropore flow in the hard pan, the lateral flux rate in the upper bund exceeded the total infiltration capacity of the hard pan. As a consequence, cross-flow occurred in three out of four experiments. This process was not only active during the short-term dye tracer experiments, but also for longer durations, where cross-flow remained stable with time for several days (data not shown).

Initial conditions before dye infiltration were shown to be different but typical for both bunds, with Bund BO being slightly wetter than Bund BY (Fig. 5). Flury et al. (1994) revealed the importance of initial water content under sprinkling irrigation, as cracks and earthworm channels conducted water only in wet plots. For ponding water conditions, how-

ever, we assume that differences are negligible. Contrarily, fast flooding in the drier Bund BY may have led to a slight overestimation of preferential flow, as most macropores were active for dye transport due to saturated conditions on top of the hard pan, while the unsaturated soil matrix was not. Adsorption of dye was presumably lower in Bund BO, especially in the topsoil, due to a low clay content (Ketelsen and Meyer-Windel, 1999). In summary, however, we assume that the disparity of observed dye patterns between both bunds originates from different bund properties and not from different initial conditions. Bund BY is characterized by a low bulk density, high infiltration rates, and enhanced macroporosity, and the main mechanisms of dye transport were convection and dispersion. In contrast, diffusion played an important role in the low-permeability Bund BO.

Strong dye coverage in the bunds reached to the maximum depth of excavation of 85 cm in all experiments, showing that solute transport in the bund toward the groundwater (groundwater level was at a depth of about 95 cm during the experiments, but is distinctly higher during the irrigation season) is of major importance. Tang et al. (2008) proposed preferential fluxes through the bunds to explain high $\text{NO}_3\text{-N}$ concentrations in wells in the same catchment, and our dye tracer data

prove that solutes actually take this pathway. Compared with the puddled paddy fields, bunds can be considered as under no-till treatment, where deeper dye penetration has been reported previously (Vervoort et al., 2001).

The data presented here are representative for the state of the bunds at the end of the rice-growing season and during reflooding in the following year. After flooding and puddling of the paddy field, the surface of the bund is plastered with soil, which may reduce infiltration until the surface will again be perforated by biopores. The shrinkage cracks in Bund BY will probably close under continuous submergence, while the biopores in both bunds will remain stable due to their cylindrical shape. We therefore assume that even newly constructed bunds will soon exhibit enhanced permeability.

CONCLUSIONS

Dye tracer tests were a suitable method to enhance knowledge about water fluxes in bunds, although dye coverage is only a semiquantitative measure (Aeby et al., 1997). The bunds surrounding paddy rice fields were shown to be distinctly more vulnerable for preferential water losses than the actual fields. The bunds thus act as a shortcut for the transport of fertilizers and pesticides applied with the irrigation water, involving the risk of direct groundwater pollution. Within the bund, macropore flow is initiated on top of the hard pan (where the soil is saturated) and may lead to an immediate and dramatic change in hydraulic conductivity of the bund on flooding. Macropores mainly consist of earthworm and cricket burrows, but also of root channels as well as shrinkage cracks in the bunds of young fields.

Solute transport in bunds can be described by three successive processes: (i) infiltration and vertical percolation in the upper bund; (ii) lateral spreading on top of the hard pan toward the adjacent field as cross-flow; and (iii) bypassing of the hard pan in preferential flow pathways partly reaching down to the groundwater. Our data contradict simulation results where no cross-flow occurred (Walker and Rushton, 1984; Huang et al., 2003), but also disprove the assumption that a hard pan would prevent underbund percolation losses (Liu et al., 2004). The dye tracer patterns show rather that high percolation rates in the bund may be associated with substantial cross-flow on top of the hard pan toward the adjacent field. The fraction of cross-flow depends on the amount of water bypassing the hard pan via macropores.

The low bulk density and associated high conductivity of young bunds increase percolation losses, while a high compaction of the bund and the prevention of macropores in the hard pan are required to increase water use efficiency; however, compaction of a bund cannot completely prevent preferential solute losses.

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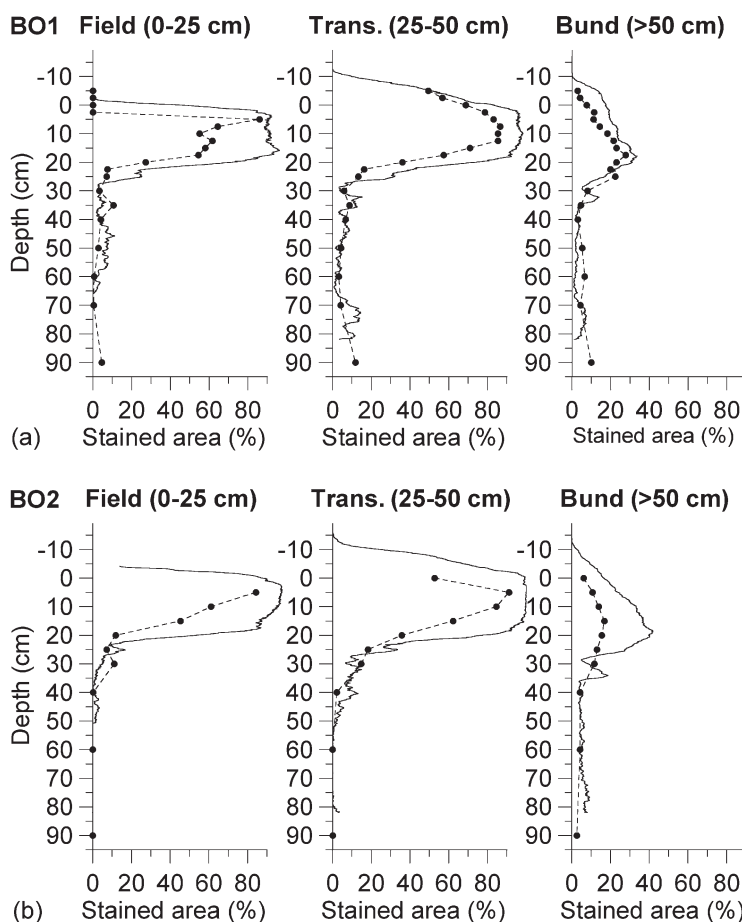


Fig. 12. Average dye coverage of Bund BO as divided into field, transition, and bund regions: (a) Exp. BO1; (b) Exp. BO2. Lines are the vertical profiles, dots are the horizontal cross-sections

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